

BUTLER ISLAND, Antarctica

WMO: 892660

Lat: 72.206S

Lon: 60.170W

Elev: 377

StdP: 14.50

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) -33.5	(c) -30.3	(d) -40.8	(e) 0.5	(f) -33.9	(g) -37.7	(h) 0.6	(i) -31.2	(j) 41.1	(k) -13.5	(l) 36.0	(m) -15.1	(n) 7.0	(o) 190	(p) 1.225

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 9.9	(c) 40.9	(d) 34.8	(e) 37.5	(f) 32.3	(g) 34.2	(h) 29.9	(i) 33.1	(j) 37.3	(k) 30.7	(l) 33.6	(m) 28.7	(n) 30.9	(o) 5.9	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.2	23.5	32.5	27.7	22.0	31.0	26.2	20.4	29.3	12.4	37.4	11.4	33.7	10.6	31.0	42.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 36.2	(b) 30.3	(c) 25.5	(d) DB	(e) -36.1	(f) 50.2	(g) 4.2	(h) 5.8	(i) -39.1	(j) 54.4	(k) -41.6	(l) 57.8	(m) -43.9	(n) 61.1	(o) -47.0	(p) 65.4	(q) N/A
(a) 36.2	(b) 30.3	(c) 25.5	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	26.1	17.1	3.1	-7.4	-9.5	-10.3	-12.6	-10.8	-5.6	0.8	16.0	25.4
	DBStd	17.26	5.41	7.97	10.74	9.55	11.03	12.12	12.29	12.56	12.57	10.20	7.15	5.18
	HDD50	17290	740	922	1452	1721	1845	1808	1939	1884	1669	1525	1021	764
	HDD65	22765	1205	1342	1917	2171	2310	2258	2404	2349	2119	1990	1471	1229
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.6	8.3	9.5	9.3	8.8	8.6	9.2	9.8	9.6	11.0	11.2	10.3	9.9
Precipitation	PrecAvg	25.8	1.5	1.8	2.0	1.9	2.3	1.9	2.3	2.6	2.8	2.8	2.4	1.5
	PrecMax	34.3	2.6	2.5	4.3	4.1	4.2	3.3	5.9	4.3	4.4	5.4	4.0	3.1
	PrecMin	17.4	0.5	0.8	0.7	0.7	0.9	0.8	1.1	0.8	1.7	1.1	1.0	0.5
	PrecStd	4.1	0.6	0.5	0.7	0.8	0.8	0.7	1.0	0.7	0.7	1.1	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.7	43.3	40.8	31.8	34.5	33.9	33.9	31.2	35.1	37.7	41.0	44.0
		MCWB	37.0	37.0	35.1	29.0	29.1	29.4	29.3	26.5	29.8	32.1	33.5	36.3
	2%	DB	40.3	37.9	33.9	19.7	26.2	25.8	25.9	25.7	29.3	28.9	34.3	38.8
		MCWB	33.8	33.0	28.9	17.4	22.4	22.4	22.5	21.4	24.8	24.1	29.2	33.3
	5%	DB	36.2	32.2	24.9	11.7	15.8	16.5	16.9	20.1	23.3	21.3	28.8	35.0
		MCWB	31.5	28.1	22.1	10.9	13.6	14.6	14.7	17.0	19.7	19.7	25.6	30.8
	10%	DB	32.8	27.8	17.6	5.5	5.6	6.3	4.3	6.9	13.4	14.4	25.7	32.0
		MCWB	29.9	26.1	16.8	4.7	5.0	5.4	3.5	6.1	11.8	13.4	23.6	29.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.9	37.0	35.5	23.4	21.1	26.4	29.4	27.2	27.6	24.6	31.3	34.3
		MCDB	40.3	42.8	40.9	27.3	24.5	30.1	33.6	31.5	32.5	28.2	35.3	39.2
	2%	WB	32.1	32.5	28.5	16.5	14.5	17.9	23.4	23.5	23.0	19.8	27.7	31.4
		MCDB	35.7	37.0	33.1	18.0	17.2	20.6	27.1	27.4	27.2	21.7	31.3	34.0
	5%	WB	30.1	28.1	22.5	11.4	6.6	9.3	16.6	19.4	16.4	14.2	24.7	29.6
		MCDB	32.4	30.8	25.5	12.3	7.1	10.5	19.1	23.2	19.0	15.7	26.5	31.6
	10%	WB	28.3	24.9	16.8	5.8	1.1	2.4	7.9	12.0	7.9	10.0	22.4	27.9
		MCDB	29.6	25.9	17.7	6.5	1.8	3.0	9.3	13.9	8.9	10.8	23.9	29.4
Mean Daily Temperature Range	5% DB	MDBR	9.9	10.2	10.5	9.8	10.5	10.7	12.0	11.7	12.4	11.7	10.7	10.0
		MCDBR	16.9	17.0	22.3	21.3	28.9	26.0	27.5	27.9	26.4	23.9	17.4	17.0
		MCWBR	10.1	14.0	16.8	17.8	25.8	23.0	23.2	23.1	22.0	19.0	12.9	11.8
	5% WB	MCDBR	11.7	14.4	20.4	20.3	18.9	21.0	26.3	26.5	25.7	19.6	12.7	11.5
		MCWBR	9.1	11.8	16.8	18.3	17.5	19.7	23.4	23.2	22.1	17.4	11.0	9.3
Clear-Sky Solar Irradiance	taub	0.256	0.242	0.211	0.138	N/A	N/A	N/A	N/A	0.140	0.208	0.228	0.241	0.255
	taud	2.242	2.266	2.261	1.987	N/A	N/A	N/A	N/A	1.934	2.201	2.226	2.246	2.233
	Ebn at Noon	309	292	261	175	0	0	0	0	162	255	296	314	315
	Edh at Noon	35	28	19	8	0	0	0	0	8	20	29	34	37
All-Sky Solar Radiation	RadAvg	2481	1607	721	148	0	0	0	0	60	470	1287	2250	2813
	RadStd	74	72	23	9	0	0	0	0	3	16	35	46	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page